COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF ENGINEERING

B. TECH DEGREE SECOND SEMESTER SECOND INTERNAL EXAMINATIONS, JULY 2024

# 23-200-0202B ENGINEERING CHEMISTRY

**(CS A &B ,EC B, IT A&B, EEE B)**

**(2023 Scheme)**

**TIME: 2 hrs** **MAX. MARKS: 30**

**Course Outcomes**

On completion of this course the student will be able to:

CO1: Explain the basic concepts of chemical thermodynamics, and quantum chemistry.

CO 2: Illustrate the spectroscopic methods in characterizing materials.

CO3: Develop electro chemical methods to protect different metals from corrosion.

CO4: Interpret the chemistry of a few important engineering materials and their industrial applications

CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations

## PART A

**(Answer all questions)** **(5 x 2= 10 Marks)**

| Q.No. | Questions | Marks | BL | CO | PO |
|---|---|---|---|---|---|
| I (a) | Why TMS is used as reference standard in NMR spectroscopy? | 2 | L2 | 2 | 3 |
| (b) | Explain the significance of fingerprint region. | 2 | L2 | 2 | 1 |
| (c) | Explain how you can calculate the bond length of a diatomic molecule from its rotational spectrum. | 2 | L3 | 2 | 1 |
| (d) | How polymers are classified based on their origin.? | 2 | L1 | 4 | 1 |
| (e) | What is electrode potential? | 2 | L1 | 3 | 1 |

## PART B

**(Answer the following questions choosing either the first OR second option)** **(2 x 10= 10 Marks)**

| Q.No. | Questions | Marks | BL | CO | PO |
|---|---|---|---|---|---|
| II | (i) Which type of nuclei are NMR active? Under what conditions such nuclei will be in resonance with radiation absorbed? | 6 | L3 | 2 | 3 |
| | (ii) The internuclear distance of CO is 1.13 Å. | 4 | L2 | 2 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| | Calculate the energy in Joules in the first excited rotational level. The atomic masses are $^{12}C = 1.99 \times 10^{-26}$ Kg and $^{16}O = 2.66 \times 10^{-26}$ Kg | | | | |
| | **OR** | | | | |
| III (a) | Explain the low resolution and high resolution NMR spectra of the following:<br>(a) $CH_3\ CH_2SH$<br>(b) $CF_2{=}CHF$<br>(c) $CH_2(CH_2CH_2Br)_2$ | 6 | L3 | 2 | 3 |
| (b) | State Beer – Lambert's law.<br>A solution of thickness 2cm transmits 40% incident light. Calculate the concentration of the solution given that $\varepsilon$ =6000 dm³/mol/cm. | 4 | L2 | 2 | 1 |
| IV (a) | Discuss different types of Lubricants and its properties. | 6 | L1 | 3 | 1 |
| (b) | Write a note on Lead storage battery. | 4 | L1 | 4 | 1 |
| | **OR** | | | | |
| V (a) | Explain the rusting of iron with the help of electrochemical theory of corrosion. | 6 | L1 | 4 | 1 |
| (b) | Write a note on classification and properties of Refractories. | 4 | L1 | 3 | 1 |

**Bloom's Taxonomy Levels**

L1 : 46.1 % L2: 30.7% L3: 23.2%